VU MECHANICS – LAWI 100515 (PI)
SS 2026

2nd Exercise Exam

26.06.2026 GROUP A

Name:	
Student ID number:	
Study program code:	
Notes	
Result	____/100

Tasks	**Possible Points**	**Achieved Points**
1) Task 1 – Section Forces and Moments	40	
2) Task 1 – Truss System	30	
3) Task 2	30	

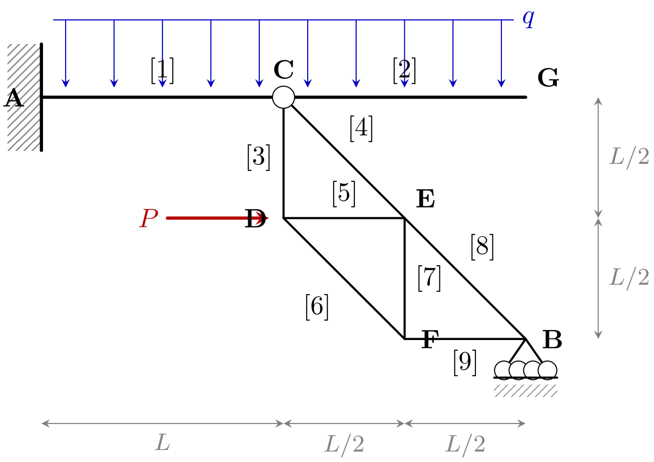
Working Time: 90 min **Passing Grade:** $\geq 50\,\%$

Permitted Aids: Formula sheet (boku-learn), calculator, set square, compass. No pencil, no red pen, no own paper.

1. Task (70 %)

Given is the planar static system shown below. Dimensions and system parameters are given in the sketch. The structure is loaded by a **uniformly distributed load** q acting on beams [1] and [2], and a **concentrated load** P at truss node D.

$L = 2\,\text{m}$	$q = 5\,\text{kN/m}$	$P = 10\,\text{kN}$



Tasks:

1. Check whether the system is statically determinate.
2. Determine the support reactions **in general form as functions of** q**,** P**, and** L. Draw the reactions into the system sketch.
3. Evaluate the support reactions for the given parameters:

   $A_V =$ $A_H =$ $M_A =$ $B_V =$

4. Determine the section forces $M(x)$, $V(x)$, and $N(x)$ in beams [1] and [2] **in general form as functions of** q**,** P**, and** L. Define the range of the running variable x. Evaluate the functions at the beam start and end:

	Beam start			Beam end		
	M [kNm]	V [kN]	N [kN]	M [kNm]	V [kN]	N [kN]
Beam [1]						
Beam [2]						

5. Draw the distributions $M(x)$, $V(x)$, and $N(x)$ and mark the characteristic values.
6. Calculate the axial forces S_8 and S_9 using a **circular cut (Rundschnitt) at node B**, in general form as a function of P:

   $S_8 =$ $S_9 =$

7. Calculate the axial forces S_4, S_5, and S_6 using **Ritter's cut (Ritterschnitt)**, in general form as a function of P:

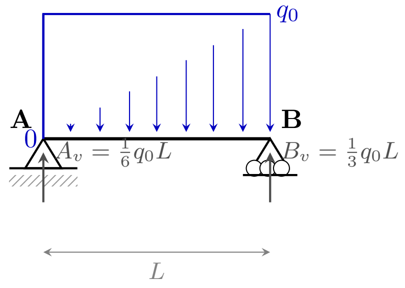
   $S_4 =$ $S_5 =$ $S_6 =$

2. Task (30 %)

Given is a simply supported beam (span L) with a triangular load (0 at A, q_0 at B). The support reactions are $A_v = \frac{1}{6}q_0 L$ and $B_v = \frac{1}{3}q_0 L$. The bending moment distribution is:

$$M(x) = -\frac{q_0}{6L}\,x^3 + \frac{q_0 L}{6}\,x \qquad (0 \le x \le L)$$

$L = 2\,\mathrm{m}$	$q_0 = 6\,\mathrm{kN/m}$



Tasks:

1. Determine the shear force $V(x)$ as a function of x.
2. Find the position $x_{\max}$ along the beam where the moment is maximum.
3. Calculate the corresponding maximum moment $M_{\max}$.
4. Sketch the distributions of $V(x)$ and $M(x)$ with all characteristic values.